

CAN THO UNIVERSITY
COLLEGE OF INFORMATION AND COMMUNICATION TECHNOLOGY
OPERATING SYSTEMS (CT104H)
LAB #1

Student name: Nguyễn Minh Nhựt

ID: B2205896

Question 1: Create a directory in your home directory called projects. In the projects directory, create nine empty files that are named house1, house2, house3, and so on to house9. Assuming there are lots of other files in that directory, come up with a single argument to ls that would list just those nine files.

Answer:

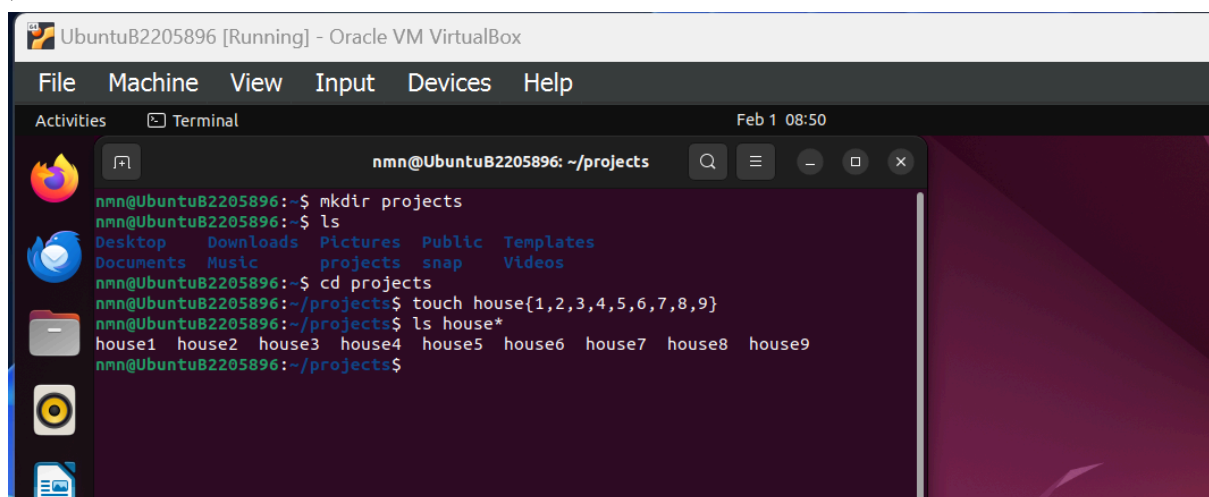
```
$mkdir projects
```

```
$ls
```

```
$cd projects
```

```
$touch house{1,2,3,4,5,6,7,8,9}
```

```
$ls house*
```



Question 2: Make the \$HOME/projects/houses/doors/ directory path. Create the following empty files within this directory path (try using absolute and relative paths from your home directory):

```
$HOME/projects/houses/bungalow.txt
```

```
$HOME/projects/houses/doors/bifold.txt
```

```
$HOME/projects/outdoors/vegetation/landscape.txt
```

Answer:

```
$cd projects
```

```
$mkdir -p $HOME/projects/houses/doors/
```

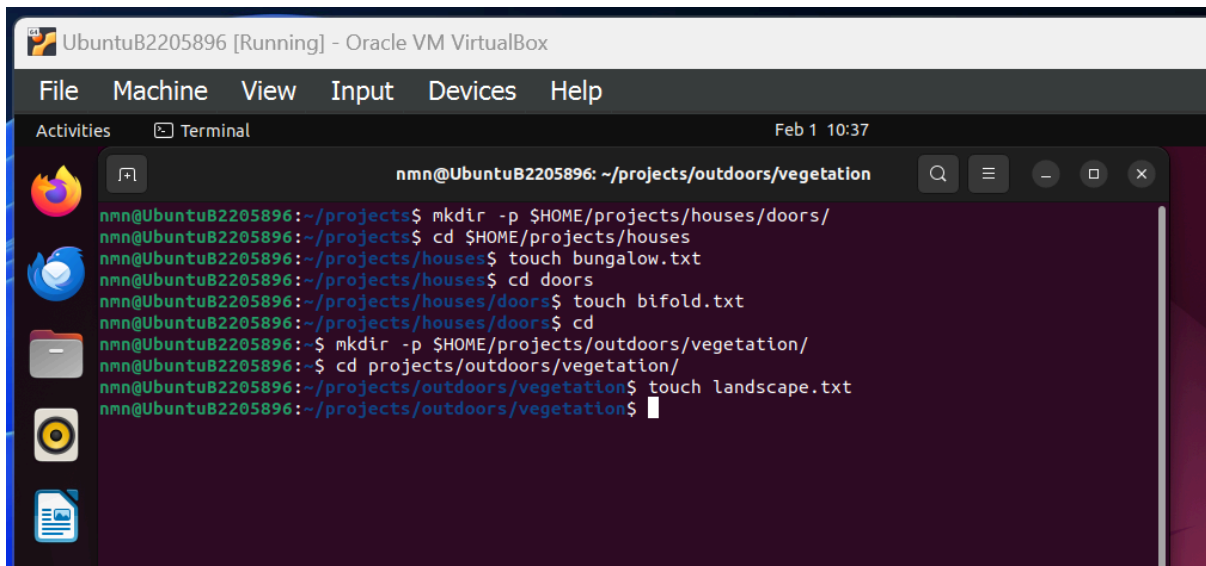
```
$cd $HOME/projects/houses
```

```
$touch bungalow.txt
```

```
$cd doors
```

```
$touch bifold.txt
```

```
$cd
$ mkdir -p $HOME/projects/outdoors/vegetation/
$cd projects/outdoors/vegetation/
$touch landscape.txt
```



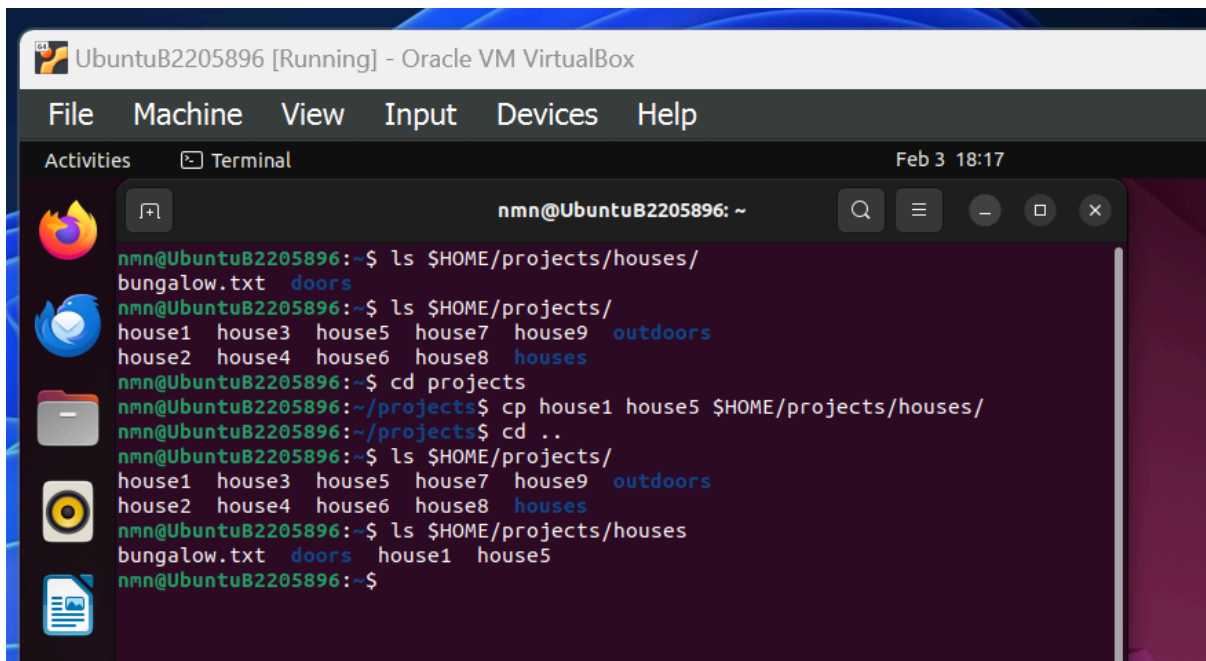
The screenshot shows a terminal window titled 'UbuntuB2205896 [Running] - Oracle VM VirtualBox'. The window has a menu bar with 'File', 'Machine', 'View', 'Input', 'Devices', and 'Help'. Below the menu bar is a toolbar with 'Activities' and 'Terminal' buttons. The terminal itself shows the following commands and output:

```
nmn@UbuntuB2205896: ~/projects/outdoors/vegetation
nmn@UbuntuB2205896:~/projects$ mkdir -p $HOME/projects/houses/doors/
nmn@UbuntuB2205896:~/projects$ cd $HOME/projects/houses
nmn@UbuntuB2205896:~/projects/houses$ touch bungalow.txt
nmn@UbuntuB2205896:~/projects/houses$ cd doors
nmn@UbuntuB2205896:~/projects/houses/doors$ touch bifold.txt
nmn@UbuntuB2205896:~/projects/houses/doors$ cd
nmn@UbuntuB2205896:~$ mkdir -p $HOME/projects/outdoors/vegetation/
nmn@UbuntuB2205896:~$ cd projects/outdoors/vegetation/
nmn@UbuntuB2205896:~/projects/outdoors/vegetation$ touch landscape.txt
nmn@UbuntuB2205896:~/projects/outdoors/vegetation$
```

Question 3: Copy the files house1 and house5 to the \$HOME/projects/houses/ directory.

Answer:

```
$ls $HOME/projects/houses/
$ls $HOME/projects/
$cd projects
$cp house1 house5 $HOME/projects/houses/
$cd ..
$ls $HOME/projects/
$ls $HOME/projects/houses/
```



The screenshot shows a terminal window titled 'UbuntuB2205896 [Running] - Oracle VM VirtualBox'. The window has a menu bar with 'File', 'Machine', 'View', 'Input', 'Devices', and 'Help'. Below the menu bar is a toolbar with 'Activities' and 'Terminal' buttons. The terminal itself shows the following commands and output:

```
nmn@UbuntuB2205896: ~
nmn@UbuntuB2205896:~$ ls $HOME/projects/houses/
bungalow.txt  doors
nmn@UbuntuB2205896:~$ ls $HOME/projects/
house1 house3 house5 house7 house9 outdoors
house2 house4 house6 house8 houses
nmn@UbuntuB2205896:~$ cd projects
nmn@UbuntuB2205896:~/projects$ cp house1 house5 $HOME/projects/houses/
nmn@UbuntuB2205896:~/projects$ cd ..
nmn@UbuntuB2205896:~$ ls $HOME/projects/
house1 house3 house5 house7 house9 outdoors
house2 house4 house6 house8 houses
nmn@UbuntuB2205896:~$ ls $HOME/projects/houses
bungalow.txt doors house1 house5
nmn@UbuntuB2205896:~$
```

Question 4: Recursively copy the /usr/share/doc/initscripts* directory to the \$HOME/projects/ directory. Maintain the current date/time stamps and permissions.

Answer:

```
$mkdir -p $HOME/share/doc/initscripts/
$cd share/doc/initscripts/
$touch nhut{1,2,3,4,5}
$cd
$cp -ra share/doc/initscripts/nhut2 $HOME/projects/
```

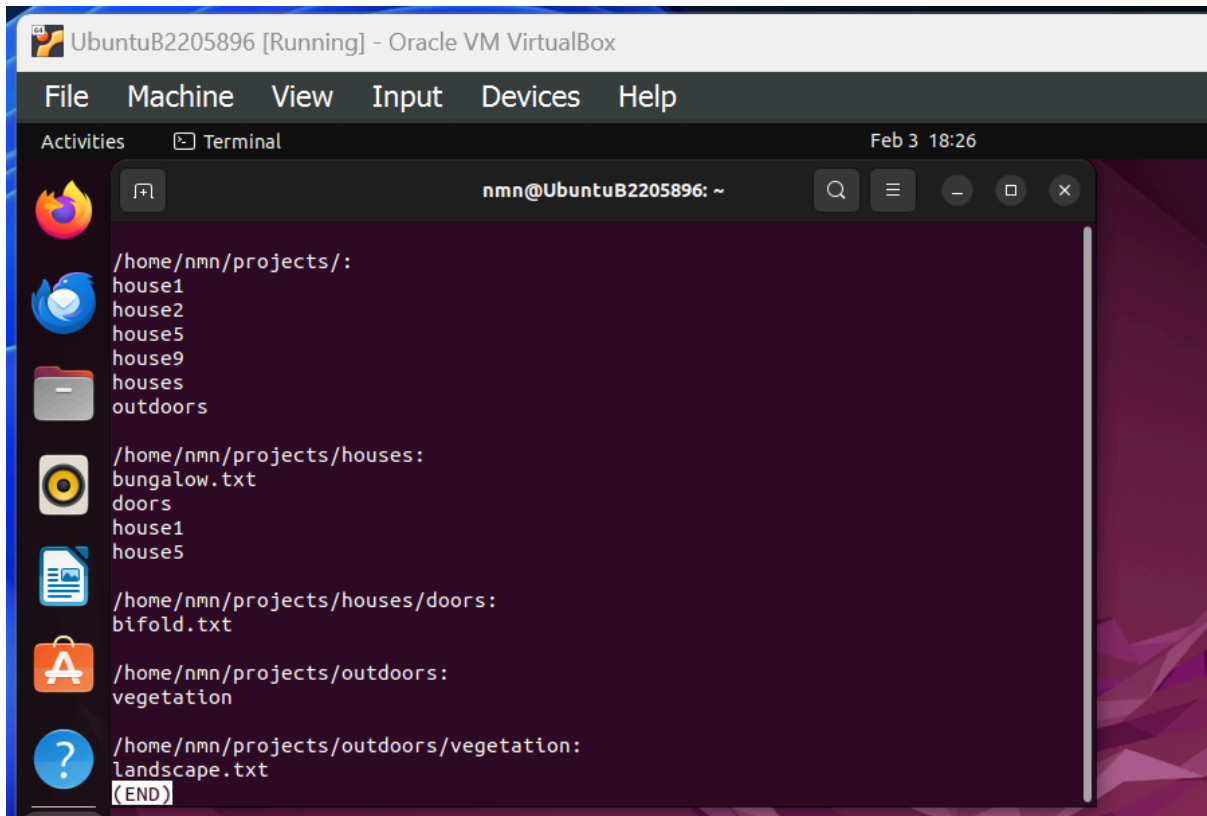
```
nmn@UbuntuB2205896: ~
nmn@UbuntuB2205896:~$ mkdir -p $HOME/share/doc/initscripts/
nmn@UbuntuB2205896:~$ cd share/doc/initscripts/
nmn@UbuntuB2205896:~/share/doc/initscripts$ touch nhut{1,2,3,4,5}
nmn@UbuntuB2205896:~/share/doc/initscripts$ cd
nmn@UbuntuB2205896:~$ cp -ra share/doc/initscripts/nhut2 $HOME/projects/
nmn@UbuntuB2205896:~$ ls $HOME/projects/
house1 house2 house3 house4 house5 house6 house7 house8 house9 houses nhut2
nmn@UbuntuB2205896:~$ ls -l $HOME/projects/
total 4
-rw-rw-r-- 1 nmn nmn 0 Feb 1 08:49 house1
-rw-rw-r-- 1 nmn nmn 0 Feb 1 08:49 house2
-rw-rw-r-- 1 nmn nmn 0 Feb 1 08:49 house3
-rw-rw-r-- 1 nmn nmn 0 Feb 1 08:49 house4
-rw-rw-r-- 1 nmn nmn 0 Feb 1 08:49 house5
-rw-rw-r-- 1 nmn nmn 0 Feb 1 08:49 house6
-rw-rw-r-- 1 nmn nmn 0 Feb 1 08:49 house7
-rw-rw-r-- 1 nmn nmn 0 Feb 1 08:49 house8
-rw-rw-r-- 1 nmn nmn 0 Feb 1 08:49 house9
drwxrwxr-x 3 nmn nmn 4096 Feb 1 09:17 houses
-rw-rw-r-- 1 nmn nmn 0 Feb 1 09:56 nhut2
nmn@UbuntuB2205896:~$
```

Question 5: Recursively list the contents of the \$HOME/projects/ directory. Pipe the output to the less command so you can page through the output.

Answer:

```
$cd
$ls -R $HOME/projects/ | less
```

```
nmn@UbuntuB2205896:~$ ls -R $HOME/projects/ | less
nmn@UbuntuB2205896:~$
```



UbuntuB2205896 [Running] - Oracle VM VirtualBox

File Machine View Input Devices Help

Activities Terminal Feb 3 18:26

nmn@UbuntuB2205896: ~

```
/home/nmn/projects/:
house1
house2
house5
house9
houses
outdoors

/home/nmn/projects/houses:
bungalow.txt
doors
house1
house5

/home/nmn/projects/houses/doors:
bifold.txt

/home/nmn/projects/outdoors:
vegetation

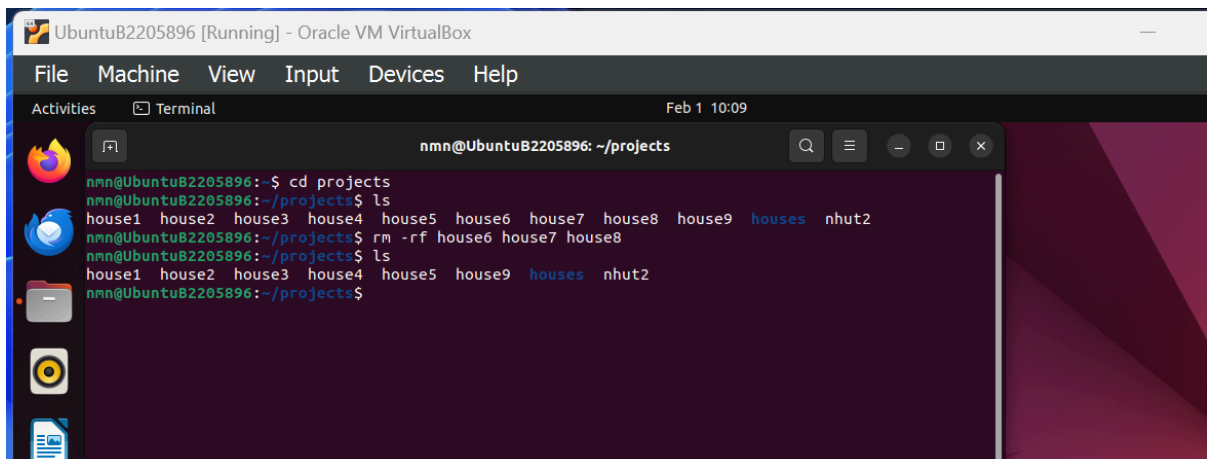
/home/nmn/projects/outdoors/vegetation:
landscape.txt
(END)
```

Question 6: Remove the files house6, house7, and house8 without being prompted.

Answer:

\$cd projects

\$rm -rf house6 house7 house 8



UbuntuB2205896 [Running] - Oracle VM VirtualBox

File Machine View Input Devices Help

Activities Terminal Feb 1 10:09

nmn@UbuntuB2205896: ~/projects

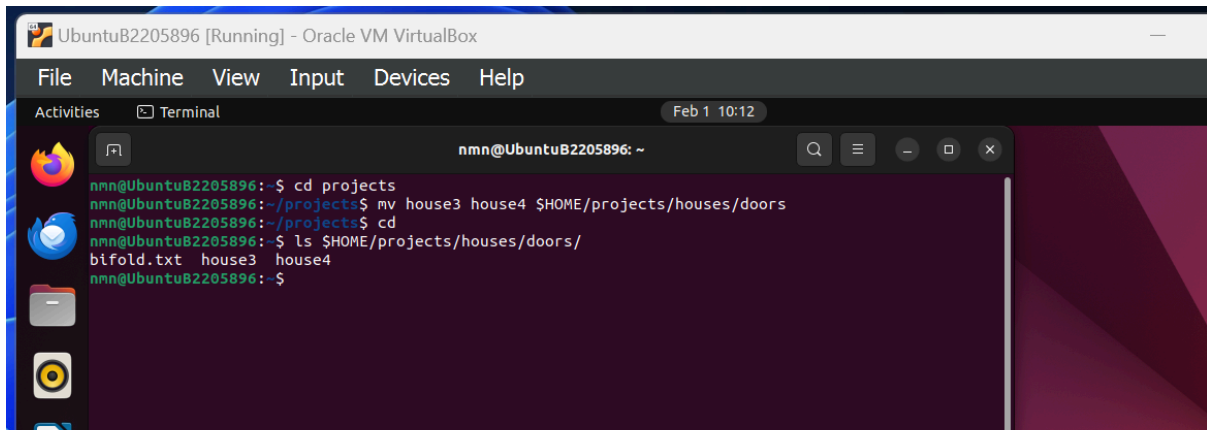
```
nmn@UbuntuB2205896:~$ cd projects
nmn@UbuntuB2205896:~/projects$ ls
house1 house2 house3 house4 house5 house6 house7 house8 house9 houses nhut2
nmn@UbuntuB2205896:~/projects$ rm -rf house6 house7 house8
nmn@UbuntuB2205896:~/projects$ ls
house1 house2 house3 house4 house5 house9 houses nhut2
nmn@UbuntuB2205896:~/projects$
```

Question 7: Move house3 and house4 to the \$HOME/projects/houses/doors directory.

Answer:

\$cd projects

\$mv house3 house4 \$HOME/projects/houses/doors



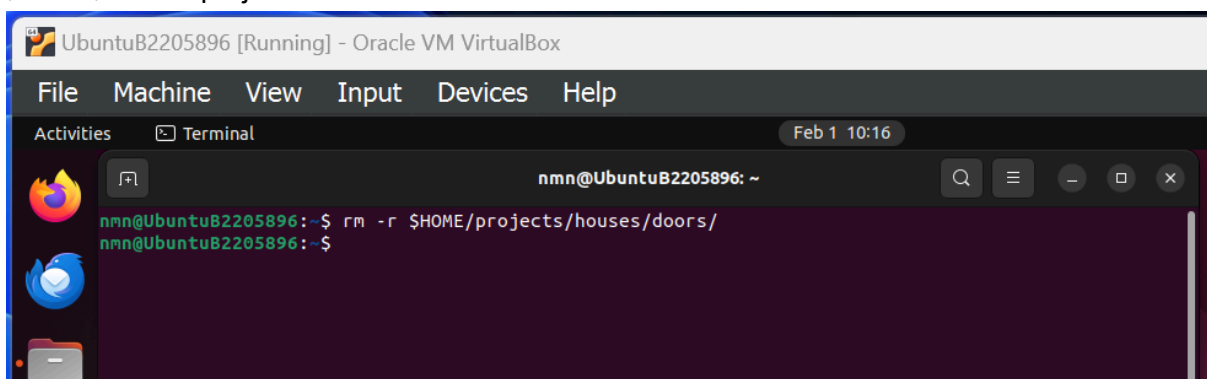
```

nmn@UbuntuB2205896:~$ cd projects
nmn@UbuntuB2205896:~/projects$ mv house3 house4 $HOME/projects/houses/doors
nmn@UbuntuB2205896:~/projects$ cd
nmn@UbuntuB2205896:~$ ls $HOME/projects/houses/doors/
bifold.txt house3 house4
nmn@UbuntuB2205896:~$
  
```

Question 8: Remove the \$HOME/projects/houses/doors directory and its contents.

Answer:

`$rm -r $HOME/projects/houses/doors/`



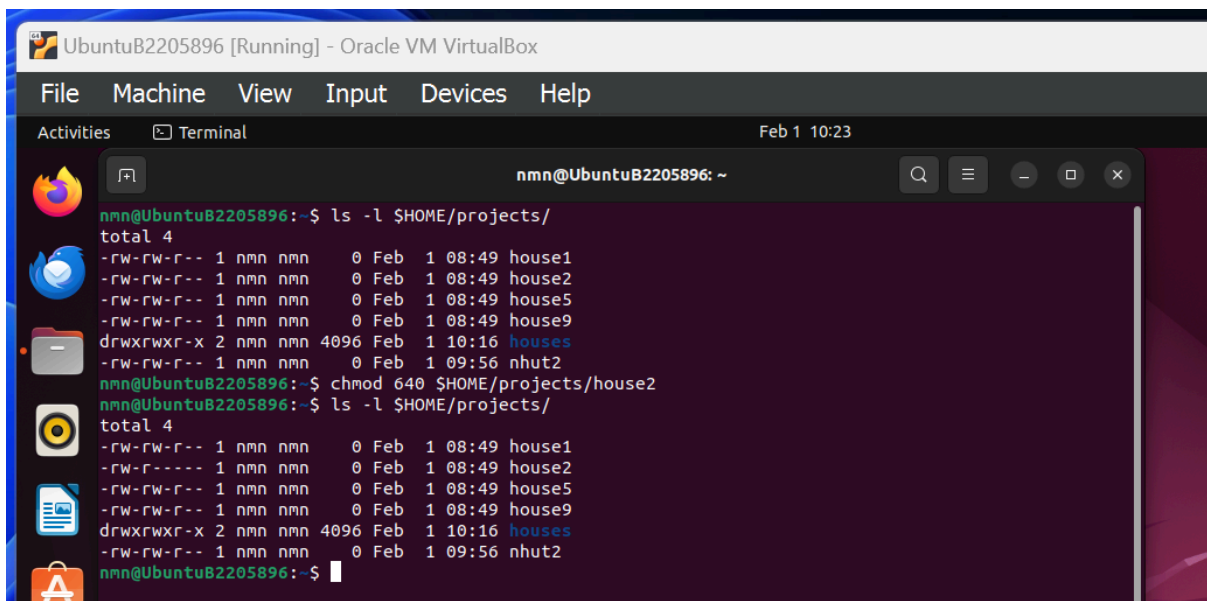
```

nmn@UbuntuB2205896:~$ rm -r $HOME/projects/houses/doors/
nmn@UbuntuB2205896:~$
  
```

Question 9: Change the permissions on the \$HOME/projects/house2 file so it can be read and written by the user who owns the file, only read by the group, and have no permission for others.

Answer:

`$chmod 640 $HOME/projects/house2`



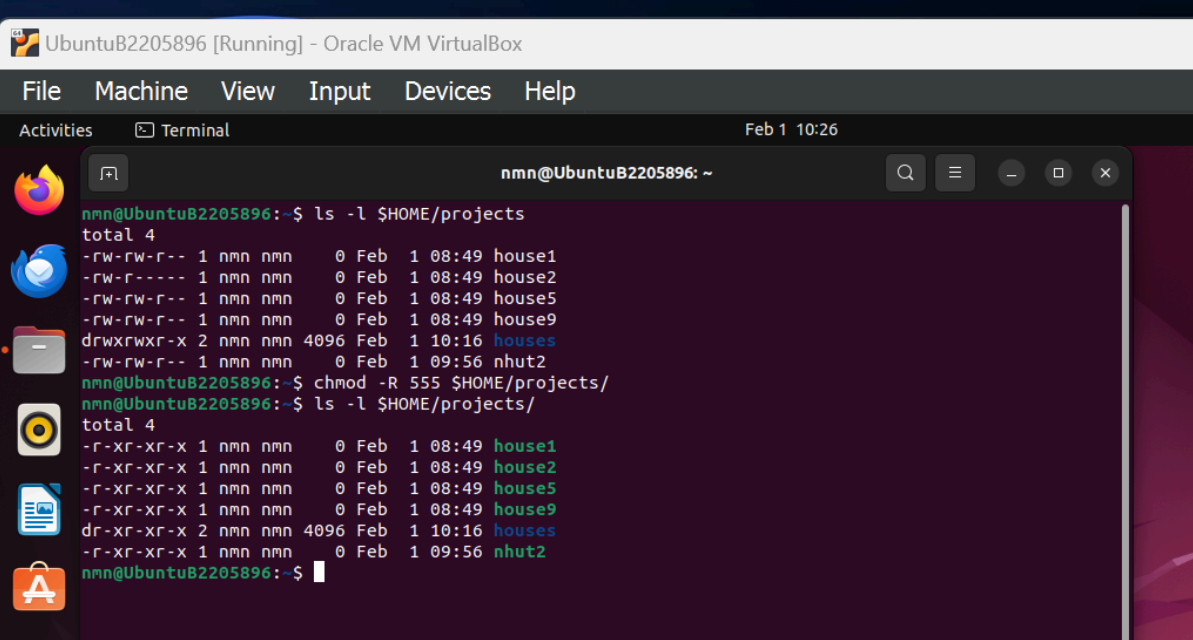
```

nmn@UbuntuB2205896:~$ ls -l $HOME/projects/
total 4
-rw-rw-r-- 1 nmn nmn 0 Feb 1 08:49 house1
-rw-rw-r-- 1 nmn nmn 0 Feb 1 08:49 house2
-rw-rw-r-- 1 nmn nmn 0 Feb 1 08:49 house5
-rw-rw-r-- 1 nmn nmn 0 Feb 1 08:49 house9
drwxrwxr-x 2 nmn nmn 4096 Feb 1 10:16 houses
-rw-rw-r-- 1 nmn nmn 0 Feb 1 09:56 nhut2
nmn@UbuntuB2205896:~$ chmod 640 $HOME/projects/house2
nmn@UbuntuB2205896:~$ ls -l $HOME/projects/
total 4
-rw-rw-r-- 1 nmn nmn 0 Feb 1 08:49 house1
-rw-r----- 1 nmn nmn 0 Feb 1 08:49 house2
-rw-rw-r-- 1 nmn nmn 0 Feb 1 08:49 house5
-rw-rw-r-- 1 nmn nmn 0 Feb 1 08:49 house9
drwxrwxr-x 2 nmn nmn 4096 Feb 1 10:16 houses
-rw-rw-r-- 1 nmn nmn 0 Feb 1 09:56 nhut2
nmn@UbuntuB2205896:~$
  
```

Question 10: Recursively change permissions of the \$HOME/projects/ directory so nobody has write permission to any files or directory beneath that point in the file system.

Answer:

`$chmod -R 555 $HOME/projects/`



The screenshot shows a terminal window titled "nmn@UbuntuB2205896: ~" with a search icon, a menu icon, and window control buttons. The terminal output is as follows:

```
nmn@UbuntuB2205896:~$ ls -l $HOME/projects
total 4
-rw-rw-r-- 1 nmn nmn 0 Feb 1 08:49 house1
-rw-r----- 1 nmn nmn 0 Feb 1 08:49 house2
-rw-rw-r-- 1 nmn nmn 0 Feb 1 08:49 house5
-rw-rw-r-- 1 nmn nmn 0 Feb 1 08:49 house9
drwxrwxr-x 2 nmn nmn 4096 Feb 1 10:16 houses
-rw-rw-r-- 1 nmn nmn 0 Feb 1 09:56 nhut2
nmn@UbuntuB2205896:~$ chmod -R 555 $HOME/projects/
nmn@UbuntuB2205896:~$ ls -l $HOME/projects/
total 4
-r-xr-xr-x 1 nmn nmn 0 Feb 1 08:49 house1
-r-xr-xr-x 1 nmn nmn 0 Feb 1 08:49 house2
-r-xr-xr-x 1 nmn nmn 0 Feb 1 08:49 house5
-r-xr-xr-x 1 nmn nmn 0 Feb 1 08:49 house9
dr-xr-xr-x 2 nmn nmn 4096 Feb 1 10:16 houses
-r-xr-xr-x 1 nmn nmn 0 Feb 1 09:56 nhut2
nmn@UbuntuB2205896:~$
```